RUBRICS FOR CASE STUDY	3	2	1
IDENTIFICATION OF PROBLEM (3)	Identifies and understands all of the main issues in the case study.	Identifies and understands some of the issues in the case study.	Identifies and understands few of the issues in the case study.
ANALYSIS OF THE ISSUES(3)	Insightful and thorough analysis of all the issues.	Superficial analysis of some of the issues.	Incomplete Analysis of the issues.
USE OF EXAMPLES(3)	Coherently integrates examples with explanations and analysis.	Explanations and analysis are mostly based on examples or other evidence.	Few to no additional example; this post is mostly a description or summary.
CONCLUSIONS/ OUTCOMES OF CASE(3)	Provides a detailed description of the outcomes of the case provides detailed and appropriate conclusions for the case.	· Provides an adequate description of the outcomes of the case. · Provides adequate and mostly appropriate conclusions for the case.	· Provides a limited description of the outcomes of the case. · Provides limited and somewhat appropriate conclusions for the case.
DELIVERY AND ENTHUSIASM(3)	Very Clear and concise flow of ideas. Demonstrate passionate interest in the topics and engagement with the class	Most ideas flow but focus is lost at times. Limited evidence of interesting and engagement with the topic	Hard to follow the flow of ideas. Lack of enthusiasm and interest.
TOTAL -15 MARKS			

DATE OF SUBMISSION FOR ASSIGNMENT -II -5 MARKS

COURSERA COMPLETION -5 MARKS Mastering Data Analysis with Pandas

https://www.coursera.org/programs/kite-it-azifx/projects/learning-path-data-analysis-with-pandas?source=search

Univariate Analysis Case Study Statements (ROLL NO FROM 1 - 34)

1. A supermarket chain analyzes the monthly revenue of its stores to determine the average earnings and seasonal variations.

2. A hospital studies the distribution of patient wait times in its emergency department to improve efficiency.

3. A bank examines the transaction amounts of customers to detect spending patterns and identify anomalies.

4. A telecom company evaluates the monthly data usage of its customers to optimize pricing plans.

5. A manufacturing plant assesses daily production output to identify trends and maintain consistency.

6. A university analyzes student exam scores to measure overall academic performance.

7. An e-commerce platform studies customer ratings for products to understand consumer satisfaction.

8. A logistics company examines delivery times across different locations to improve service quality.

9. A clothing brand evaluates the monthly sales volume of a popular product to forecast future demand.

10. A hotel chain studies the average occupancy rate per month to optimize pricing strategies.

11. A government agency analyzes the age distribution of its employees for workforce planning.

12. A cinema evaluates the number of tickets sold per showtime to optimize movie scheduling.

13. A fitness center studies the average duration of gym visits to design workout programs.

14. A real estate company analyzes rental prices in different neighborhoods to determine price trends.

15. A social media platform examines daily active user counts to monitor engagement.

16. A sports league studies the number of goals scored per match to assess player performance.

17. A tech company analyzes software bug reports per month to measure system stability.

18. A coffee shop chain evaluates the number of customers visiting per day to optimize staffing.

19. A car rental service studies the average rental duration of vehicles to refine pricing models.

20. A pharmaceutical company examines daily medicine production to ensure supply meets demand.

21. A university studies the number of books borrowed per day from the library to enhance resource allocation.

22. A travel agency evaluates the average vacation package cost to understand customer spending behavior.

23. A call center examines call durations to improve customer service efficiency.

24. A food delivery service studies the average delivery time to maintain customer satisfaction.

25. A law firm evaluates the number of cases handled by each lawyer to assess workload distribution.

26. A cybersecurity firm analyzes the number of security breaches reported per week to strengthen defenses.

27. A museum examines visitor footfall per day to optimize opening hours.

28. A construction company studies the cost of materials per project to control expenses.

29. A publishing house analyzes the number of books sold per month to measure author performance.

30. A streaming platform studies average watch times per user to enhance content recommendations.

31. A government agency examines water consumption per household to promote conservation efforts.

32. A taxi service evaluates the average number of rides per driver to measure efficiency.

33. A gaming company studies daily login counts of online players to track engagement levels.

34. A financial firm analyzes customer credit scores to assess lending risk.

Steps for Performing Univariate Analysis on a Dataset

Univariate analysis focuses on examining a single variable in a dataset to understand its distribution, central tendency, and variability. Below are the key steps:

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1. Define the Objective

- Identify the variable to be analyzed (e.g., sales, age, income).
- Determine the purpose of the analysis (e.g., understanding distribution, detecting outliers, summarizing data).

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2. Data Collection & Preprocessing

- **Load the dataset** (from CSV, database, API, etc.).
- **Check for missing values** and handle them (e.g., imputation or removal).
- **Convert data types** (if necessary, e.g., numeric to categorical).
- **Remove duplicates** to ensure clean data.

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3. Descriptive Statistics

- **Measure Central Tendency**:
  - **Mean**: Average value of the variable.
  - **Median**: Middle value in an ordered dataset.
  - **Mode**: Most frequently occurring value.
- **Measure Dispersion (Variability):**
  - **Range**: Difference between max and min values.
  - **Variance & Standard Deviation**: Measure spread of data.
  - **Interquartile Range (IQR)**: Helps detect outliers.

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4. Data Visualization

- **Histogram**: Shows the frequency distribution of the variable.
- **Box Plot**: Identifies outliers and spread of data.
- **Bar Chart**: Useful for categorical data.
- **Density Plot**: Displays probability distribution.

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5. Outlier Detection & Handling

- Identify extreme values using:
  - **Box plot (IQR method):** Values outside 1.5×IQR are considered outliers.
  - **Z-score method:** If a value's z-score > 3 or < -3, it is an outlier.
- Handle outliers by:
  - Removing them (if erroneous data).
  - Transforming the data (e.g., log transformation).
  - Using robust statistical methods.

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6. Data Transformation (if needed)

- **Normalization**: Scale values to a standard range (e.g., 0 to 1).
- **Standardization**: Convert data to have a mean of 0 and standard deviation of 1.
- **Log Transformation**: Used for skewed data.

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7. Interpret Findings & Draw Conclusions

- Summarize key statistics and distribution.
- Identify trends or anomalies.
- Provide insights for decision-making.

Bivariate Analysis Case Study Statements (ROLL NO FROM 34-68)

1. A retail store examines the relationship between advertising spend and monthly sales to optimize marketing budgets.

2. A hospital investigates the correlation between patient age and recovery time to improve treatment plans.

3. A bank studies how income levels affect loan approval rates to refine lending criteria.

4. A telecom provider analyzes the impact of internet speed on customer satisfaction scores.

5. A manufacturing company explores the relationship between machine maintenance frequency and defect rates.

6. A university examines the effect of class attendance on student grades.

7. An e-commerce platform analyzes how product price influences customer purchase decisions.

8. A logistics company studies the correlation between delivery distance and fuel consumption.

9. A clothing brand examines the impact of social media campaigns on sales growth.

10. A hotel chain investigates the relationship between customer ratings and repeat bookings.

11. A government agency studies the effect of pollution levels on respiratory illness cases.

12. A fitness center explores the correlation between workout frequency and weight loss results.

13. A real estate firm analyzes how property size affects rental prices.

14. A social media company examines the connection between engagement metrics (likes, shares) and ad revenue.

15. A sports team investigates how player salaries impact team performance.

16. A tech firm examines the relationship between software update frequency and customer retention.

17. A coffee shop chain analyzes the effect of seasonal promotions on foot traffic.

18. A car rental service studies the correlation between rental price and customer satisfaction.
19. A pharmaceutical company investigates how drug dosage affects patient recovery speed.
20. A university explores the impact of study hours on final exam scores.
21. A travel agency examines how travel package discounts affect booking rates.
22. A call center investigates the correlation between customer wait time and service satisfaction.
23. A food delivery app studies how delivery time impacts customer ratings.
24. A law firm explores the effect of case complexity on resolution time.
25. A cybersecurity firm examines the relationship between system updates and vulnerability detection.
26. A museum studies how ticket prices affect daily visitor counts.
27. A construction company investigates the correlation between project deadlines and labor costs.
28. A publishing house analyzes how book reviews impact sales.
29. A streaming service explores the connection between subscription fees and user retention.
30. A government agency examines the relationship between public transportation fares and commuter usage.
31. A taxi service studies the correlation between ride prices and customer loyalty.
32. A gaming company investigates how in-game purchases impact player engagement.
33. A financial firm examines the relationship between interest rates and loan repayment rates.
34. A restaurant chain studies the effect of menu pricing on customer spending habits.

Steps for Performing Bivariate Analysis on a Dataset

Bivariate analysis examines the relationship between two variables in a dataset to identify patterns, correlations, and dependencies. It helps in understanding how one variable influences another.

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1. Define the Objective

- Identify the two variables to analyze (e.g., "Sales vs. Advertising Spend").
- Determine the purpose:
  - Are you checking for correlation?
  - Are you trying to predict one variable based on another?
  - Are you exploring trends and dependencies?

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2. Data Collection & Preprocessing

- **Load the dataset** (CSV, database, API, etc.).
- **Handle missing values** (e.g., drop, impute).
- **Convert data types** (e.g., categorical to numerical if needed).
- **Check for duplicate and inconsistent values.**
- **Ensure both variables are properly formatted and usable.**

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3. Identify the Type of Variables

The type of variables determines the method of analysis:

- **Numerical vs. Numerical:** Correlation, scatter plots, regression analysis.
- **Numerical vs. Categorical:** Box plots, group comparisons, t-tests.
- **Categorical vs. Categorical:** Crosstabs, chi-square test, stacked bar charts.

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4. Descriptive Statistics for Both Variables

- **Measure Central Tendency (Mean, Median, Mode)** for numerical variables.
- **Calculate Variability (Standard Deviation, Range, IQR).**
- **Check frequency distributions for categorical variables.**
- **Use correlation coefficient for numerical variables.**

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5. Data Visualization for Relationship Analysis

For Numerical vs. Numerical:

- **Scatter Plot**: Shows the relationship between two numerical variables.
- **Correlation Heatmap**: Displays correlation strength between variables.
- **Line Chart**: Useful if data is time-dependent.

For Numerical vs. Categorical:

- **Box Plot**: Shows distribution of numerical data across categories.
- **Violin Plot**: Similar to a box plot but includes density estimation.

For Categorical vs. Categorical:

- **Stacked Bar Chart**: Shows category proportions.
- **Chi-Square Test**: Measures association between two categorical variables.

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6. Statistical Tests to Measure Relationship Strength

For Numerical vs. Numerical:

- **Pearson Correlation Coefficient (r):** Measures linear correlation.

- **Spearman Correlation:** Used for non-linear relationships.
- **Regression Analysis:** Determines the dependency of one variable on another.

For Numerical vs. Categorical:

- **T-test:** Compares means between two groups.
- **ANOVA (Analysis of Variance):** Compares means across multiple groups.

For Categorical vs. Categorical:

- **Chi-Square Test:** Checks if two categorical variables are related.

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7. Interpretation of Results

- Identify if there is a strong, weak, or no relationship.
- Determine whether the relationship is **positive, negative, or neutral**.
- Use statistical tests to validate findings.